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Large scale study on PFASs levels in fruits, vegetables and soil from allotments and gardens contaminated by atmospheric deposition from a Dutch fluorochemical production plant

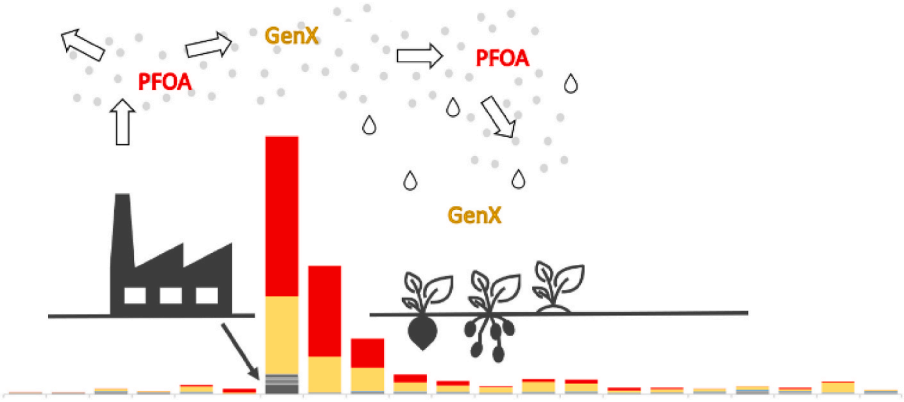
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HIGHLIGHTS

- PFASs analysed in 737 samples of garden produce close to a fluorochemical plant.
- PFOA and GenX were the predominantly observed PFASs.
- Levels were highest directly downwind of the fluorochemical production plant.
- Levels of PFASs were highest in brassicas, leafy vegetables and root vegetables.
- BAFs for PFOA in this study were 10-fold lower than those reported in literature.

GRAPHICAL ABSTRACT



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ABSTRACT

Citizens grow their own fruits and vegetables in allotment gardens in the vicinity of a fluorochemical production plant (FCPP) in The Netherlands. Historic emissions and the subsequent atmospheric deposition of perfluorooctanoic acid (PFOA) and GenX (hexafluoropropylene oxide-dimer acid/HFPO-DA) from the FCPP have resulted in the nearby environment being contaminated with per- and polyfluoroalkyl substances (PFASs). This research aimed to investigate the levels of PFASs in garden produce and whether a gradient can be observed in relation to distance from the FCPP. Furthermore, differences between certain types of fruits and vegetables were explored, as well as a potential relation between the measured concentrations in garden produce and soil. 737 fruit and vegetable samples were collected from 17 allotments and 4 gardens up to 20 km from the FCPP, along with soil and water samples. Garden produce included fruits, potatoes, fruiting vegetables, brassicas, leafy

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vegetables, root vegetables, bulb vegetables, legumes and stem vegetables. PFASs concentrations in the samples were quantified using a very sensitive UPLC-MS/MS method. PFASs were detected in most samples above the analytical limit of detection (0.3–12.5 pg/g ww). PFOA and GenX were found in the highest concentrations (up to 5280 pg/g ww GenX and 3020 pg/g ww PFOA) in garden produce sampled downwind and close to the FCPP. Other PFASs were also found, but at (much) lower levels. Field-derived bioaccumulation factors (BAFs) were calculated for PFOA and GenX. The BAFs for PFOA were shown to be approximately 1 order of magnitude lower than BAFs from other studies. This may be explained by aging of the PFASs contamination and the intense cultivation of the garden plots. This study shows that PFOA and GenX can end up in garden produce and this will result in human exposure when the garden produce is consumed.

1. Introduction

Poly- and perfluoroalkyl substances (PFASs) are a group of persistent chemicals that are ubiquitously found in the environment and have been proven to end up in the human food chain. PFASs consist of a large number of compounds (OECD, 2021) and have been produced for numerous consumer, societal and industrial applications (Buck et al., 2011). PFASs have gained a great deal of political and scientific attention over the last decade. The European Food Safety Authority (EFSA) performed a risk assessment for PFASs from exposure through food. Specifically, based on immunotoxic effects observed in children, EFSA derived a health-based guidance value for the 4 individual PFASs detected in the blood of most of the children, being PFOA (perfluorooctanoic acid), PFNA (perfluorononanoic acid), PFHxS (perfluorohexane sulphonic acid) and PFOS (perfluorooctane sulphonic acid). The established value is a tolerable weekly intake (TWI) of 4.4 ng/kg bw/day for the sum of the 4 aforementioned individual PFASs (EFSA, 2020).

Human exposure to PFASs comes from many different exposure pathways. Dietary exposure mainly takes place through the consumption of tap water and food (EFSA et al., 2020). The average current dietary exposure to PFOA, PFOS, PFHxS and PFNA exceeds the TWI for toddlers, young children and for large proportions of other population groups (adolescents, adults, elderly) in Europe (EFSA et al., 2020). Considering the low TWI, analytical methodologies able to detect very low levels in food and tap water are required to provide meaningful exposure assessments. This was demonstrated by the high upper bound exposure estimated by EFSA, based on LOQs for left-censored data. Although very sensitive techniques have been reported (Ballesteros-Gómez et al., 2010; Berendsen et al., 2020), methods specifically focussing on a wide array of fruits and vegetables (garden produce) are scarce. Recently, a method has been published that enables a (sub) pg/g detection of PFASs in garden produce (Kause et al., 2024), and the same method has been applied in this investigation.

The fluorochemical production plant (FCPP) located in the vicinity of Dordrecht, the Netherlands, has emitted PFOA and GenX (also known as FRD-903/hexafluoropropylene oxide-dimer acid/HFPO-DA/2,3,3,3-tetrafluoro-2-(heptafluoropropoxy)propanoic acid) for decades, resulting in the contamination of the nearby aquatic, aerial and terrestrial environments. The presence of PFOA, GenX and other PFASs was shown in soil, surface water, groundwater and leaves (Beekman et al., 2016; Zeilmaker et al., 2016; Brandsma et al., 2019; Gebbink and van Leeuwen, 2020; van Bentum and Pancras, 2023). PFOA was used as a processing agent in the production of PTFE (polytetrafluoroethylene) and FEP (fluorinated ethylene propylene) at the site up to 2012, when the use of PFOA was phased out and the FCPP switched to the use of GenX (Zeilmaker et al., 2016). The recorded emissions of PFOA and GenX between 1985 and 2015 are shown in the supplemental information (Fig. S1). PFOA was emitted at approximately 5 tonnes/year from 1985 to 2000 (Koch et al., 2017). This reduced to less than 500 kg/year between 2003 and 2012 (Zeilmaker et al., 2016). GenX was then emitted at 200–400 kg/year between 2012 and 2017 (Beekman et al., 2016). The permit for GenX and PFOA at the site has been updated and from 2022, the permitted amounts of PFOA and GenX have been reduced to 0.15 and 4 kg/year, respectively (DCMR, 2023).

Many citizens have allotments where they grow their own fruits and vegetables, and it can be perceived that citizens who have allotments in the close vicinity of the FCPP are concerned about the effects of PFASs exposure from the consumption of locally produced food on their health. The risk of eating produce from these gardens was evaluated in 2017 (Mengelers et al., 2018), for gardens close to the FCPP, a negative advice was given for consumption of their garden produce. However, given the decrease in the TWI to 4.4 ng/kg bw/week, there is a need to understand better whether the fruits and vegetables from the allotment gardens further away from the FCPP also pose a risk to consumers, and if measures need to be taken to mitigate these risks. Owners/users of contaminated allotment gardens are considered as the part of the population with the highest risk of exposure to PFASs, as a high portion of their daily diet derives from the fruits and vegetables grown in their contaminated garden.

Over the last decade, substantial data became available demonstrating that PFAS can be taken up by (edible) plants. Reviews on the uptake, translocation, accumulation and determining factors have been published in recent years (Ghisi et al., 2019; Lesmeister et al., 2021; Mei et al., 2021; Xu et al., 2022; Adu et al., 2023). The uptake of PFASs by produce depends on the part of the plant that is considered, the plant species and the type of PFAS (Costello and Lee, 2020; Mei et al., 2021). Initial root uptake of PFASs differs per plant and is PFAS dependent (Blaine et al., 2014). Whereas longer chain PFASs primarily occur in the roots, shorter chain PFASs are preferentially transported through the xylem or phloem towards shoots and fruits. Little is known on the occurrence of a variety of PFASs in edible garden produce under field conditions in allotment gardens. When PFAS data in garden produce are lacking, bioaccumulation factors (BAFs) to soil provide an estimate of the uptake of PFASs by fruits and vegetables. Overviews of derived BAFs have been given in several review papers.(Ghisi et al., 2019; Costello and Lee, 2020; Lesmeister et al., 2021; Xu et al., 2022). However, most derived BAFs originate from controlled experiments. Field-derived bioaccumulation factors are scarce, especially for locations near fluorochemical plants where historical air deposition of PFASs has occurred, and where contributions of air deposition during sampling do not confound the derivation of plant-soil bioaccumulation factors (Costello and Lee, 2020). At the current FCPP, PFOA air deposition stopped approximately 9 years ago which means that field derived BAFs can be calculated in the weathered soils.

In order to fill the knowledge gaps, a study was performed to quantify the concentrations of PFASs in fruits and vegetables from allotment gardens in close vicinity of the FCPP, to be used for a new risk assessment. The study was designed to address the following questions: (i) what are the levels of PFASs (and the PFASs profile) in garden produce; (ii) can a gradient be observed in PFASs levels in relation to distance and position of the allotments and gardens in relation to the FCPP, (iii) do certain fruits and vegetables stand out in terms of PFASs contamination and (iv) can the PFASs concentrations in the garden produce be explained by the soil concentrations? Here we report on the results related to these questions. This data from the current study was also used to update the 2017 risk assessment (Mengelers et al., 2018). That updated risk assessment, using relative potency factors (Bil et al.,

2021), showed that consumers eating fruits and vegetables from allotments and gardens closest (and primarily downwind) the FCPP were exposed to too high PFASs-levels. This risk assessment prompted RIVM to advise to limit or omit the consumption of citizen's own produce. At larger distance from the FCPP, it was safe to consume the fruits and vegetables. The details of that risk assessment have been reported elsewhere (Boon and te Biesebeek, 2022).

2. Materials & methods

2.1. Sampling area description

Allotments in this study are defined as an area of land where several individuals have been assigned a plot of soil (garden) for their own gardening of fruits and vegetables. These allotments, owned by the municipality, comprised multiple (20–100) gardens. There were 17 allotments included in this study. In addition to the allotments, 3 privately owned gardens (i.e. individual gardens) close to the FCPP, and one located approximately 10 km downwind from the FCPP were also included. The allotments and gardens were selected for the investigation based on the prevailing wind direction (south-west to north-east), distance to the FCPP and previously detected elevated PFASs levels in the soil compared to (ambient) background levels (Provincie Zuid-Holland, 2024). Seven allotments and 4 private gardens were selected on a transect from south-west to north-east, from 9 km upwind to 14 km downwind. Eight allotments were located lateral to the wind direction. Two allotments further upwind (13–20 km), from an area with low ambient background PFASs concentrations in soil, were selected as reference locations (see Fig. 1). The allotments were divided into sections in which multiple gardens were clustered, to be able to compare soil and garden produce samples.

2.2. Garden produce sampling and chemical analysis

Garden produce samples were taken from July to December 2021. From each allotment between 19 and 25 distinct types of garden produce (fruits and vegetables) were sampled, based on availability. An additional selection criterium was that the type of fruit and vegetable is substantially consumed according to the Dutch national food consumption survey (RIVM, 2020). Nine categories were defined; fruits, potatoes, fruiting vegetables, brassicas, leafy vegetables, root vegetables, bulb vegetables, legumes and stem vegetables. Within each garden produce category between 2 and 5 distinct types of garden produce were collected. An overview of the garden produce sampled is given in Table S1 in the supplemental information. Approximately 100 g of garden produce were collected per sample and they were placed in polyethylene bags. In total about 1500 individual samples were collected and samples were sent to the laboratory of Wageningen Food Safety Research on the day of collection. In the laboratory, the samples were washed, and the edible part was taken for pooling. Pumpkin, potato, beetroot, carrot and onion were all peeled. For 9 potato, 12 beetroot and 13 carrots, both peeled and unpeeled produce were analysed. When assessing the data of peeled vs unpeeled produce, the results were inconclusive (van Bentum and Pancras, 2023), and therefore the effect of peeling will not be further discussed in this paper. None of the garden produce were cooked or prepared otherwise. 1 to 3 samples of the same type of vegetable or fruit from a section within the allotment were pooled together and homogenised by grinding. In total, 737 pooled samples were analysed.

The PFASs in this study included perfluorobutanoic acid (PFBA), perfluoropentanoic acid (PFPeA), perfluorohexanoic acid (PFHxA), perfluoroheptanoic acid (PFHpA), perfluorooctanoic acid (PFOA), perfluorononanoic acid (PFNA), perfluorodecanoic acid (PFDA), perfluoroundecanoic acid (PFUnDA), perfluorododecanoic acid (PFDoDA), perfluorotridecanoic acid (PFTrDA), and perfluorotetradecanoic acid

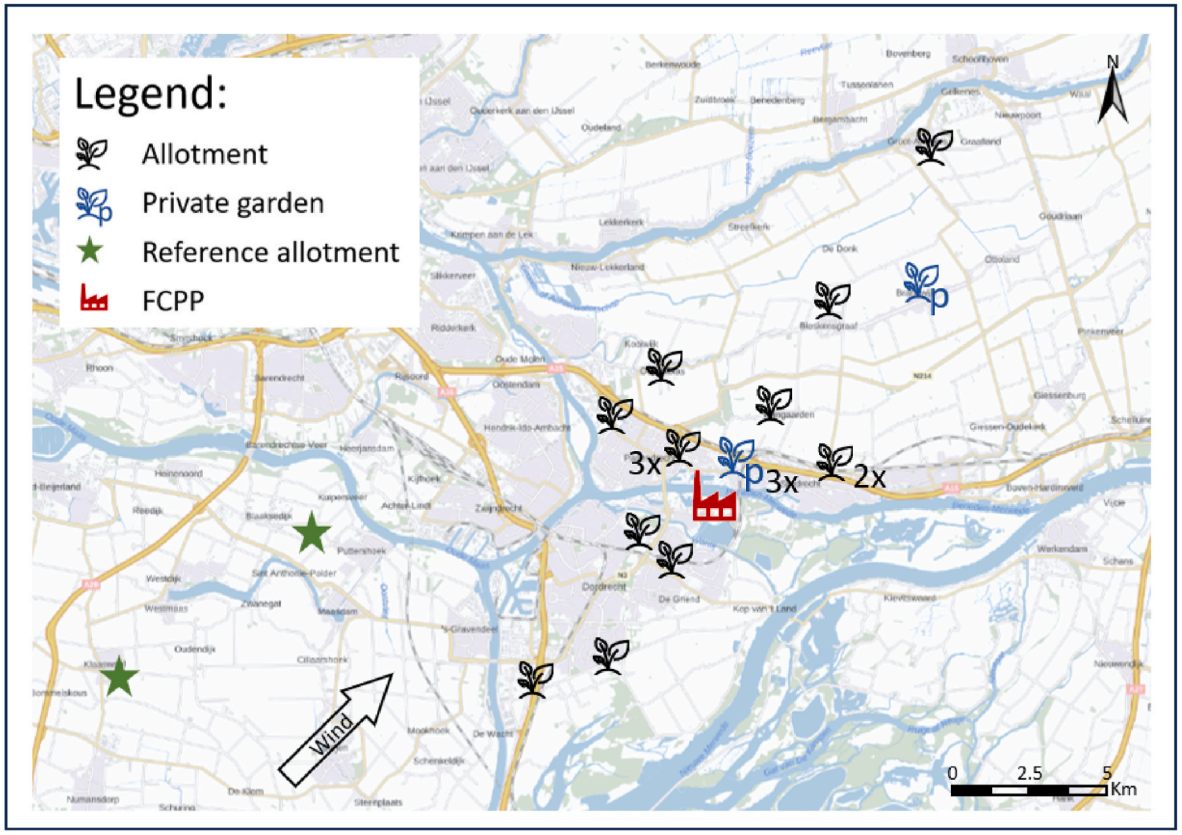


Fig. 1. Overview of the location of the FCPP and the allotments. The predominant wind direction is from South-West to North-East.

(PFTeDA), perfluorobutane sulphonic acid (PFBS), perfluorohexane sulphonic acid (PFHxS), perfluoroheptane sulphonic acid (PFHpS), perfluorooctane sulphonic acid (PFOS) and perfluorodecane sulfonic acid (PFDS), hexafluoropropylene oxide-dimer acid (HFPO-DA), also known as GenX. All compounds were analysed and reported in their anionic form. The isotopically labelled compounds used as internal standards (IS) are given in the Supplemental Information. The chemical analysis was performed using a very sensitive analytical approach, capable of reaching low limits of detection (LOD) ranging from 0.25 to 12.5 pg/g ww with a median value of 0.5 pg/g ww. The details of the method are published elsewhere (Kause et al., 2024) and can be found in the supplemental information, paragraph S1 and Table S2. In brief, 10 ± 0.5 g of the homogenised fresh sample was weighed, internal standard was added (50 μl of a 1 ng/ml solution) and the sample was extracted with 15 ml of methanol. After centrifugation, the supernatant was taken to clean-up by solid phase extraction (SPE) on a weak anion exchange column (Strata-X-AW Polymeric WAX 200mg/6 mL, 33 μm; Phenomenex, Torrance, USA). The PFASs were eluted and the extract was analysed by UPLC-MS/MS analysis. The UPLC-MS/MS analysis was performed using a Sciex ExionLC UPLC system (Sciex, Framingham, MA, USA), using a C18 reversed phase column, connected to a Sciex QTRAP 7500 system in negative electrospray ionization (ESI-) mode. Confirmation of identity of each PFAS was based on retention time match with internal standard and ion ratio of qualifier and quantifier ion. The method was validated and ISO-17025 accredited. QA/QC was crucial during the analysis of these samples. Measures to control the quality of the analysis included (i) the use of multiple mass labelled internal standards spiked to every sample (ii) the analysis of procedural blanks covering the whole sample extraction and clean-up (iii) the analysis of blank LC-MS/MS runs, (iv) the analysis of a spiked sample at two levels (2.5 and 50 pg/g ww), (v) acceptability criteria for IS recovery and accuracy, (vi) criteria for deviation of ion ratio (maximum 30%). All these measures were applied to every batch of samples analysed in this study. The chemical analysis of PFBA showed high blank levels, and therefore no meaningful results could be reported. This was also the case for a part of the PFPeA results (38% of the samples could not be assessed). The LOQ was determined as the lowest level of the calibration line where the signal to noise ratio was 6. In case there was a trace of a compound detected in the procedural blank (in some cases for PFOA), the LOQ was determined as 3.3 times the average of the response factors of the procedural blanks and converted to the corresponding concentration (Kause et al., 2024). Data on the performance of the method can be found elsewhere (Kause et al., 2024).

2.3. Garden soil sampling and chemical analysis

Soil samples were taken in the first six months of 2022, within each section where garden produce was collected. Soil samples were taken at different depths from multiple locations. Of these samples mixed bulk samples (157) were made for two depths (layer I; 0–20 cm deep, layer II; 20–50 cm deep) to represent the average concentration across both soil depths of the section. All sampling was done according to the relevant Dutch BRL methods (SIKB, 2018a, b) by a certified company (Tritium Advies B.V.). Soils were analysed for dry weight (DW), organic matter (OM) and clay fraction (as <2 μm) according to a standard method (AS3000) (SIKB, 2020) and the standard NEN-EN 15934:2012 (NEN, 2012) by the laboratory AL-West. PFASs concentrations were determined according to DIN 38414–14:2011–08 (DIN, 2010) with some minor deviations by AL-West and expressed as μg/kg dry matter. Of the homogenised, dried (40 °C) and ground sample, 1 ± 0.1 g was weighed, internal standards were added and extracted with 5 ml of methanol by orbital shaking at 225 rpm for 1 h. After centrifugation, the supernatant was analysed by LC-ESI-MS/MS. Confirmation of identity of each PFAS was based on retention time match with internal standard and ion ratio of qualifier and quantifier ion. An overview of the different measured PFASs in soil is given in the SI, Table S3.

Irrigation water, surface water and groundwater were also sampled for PFASs. Levels of PFASs in the water used for irrigation were low, and therefore not reported here. The water data is reported elsewhere (van Bentum and Pancras, 2023).

3. Results & discussion

3.1. Levels of PFASs in garden produce

The PFASs analytical results in fruits and vegetables are reported on wet weight (ww) basis (a.k.a. product basis or fresh weight basis), as commonly done in the food safety domain. Although a wide range of PFASs could be detected in the samples, the predominant compounds observed were PFOA and GenX. Average concentrations across all different types of garden produce ranged from 4 to 400 pg/g ww for PFOA and <0.25–810 pg/g ww for GenX. Averages for the sum of the other PFASs were between <0.25 and 100 pg/g ww. Of the sum of PFASs measured (setting values < LOQ to zero, i.e. lowerbound), on average and on a weight basis, PFOA made up 40%, and GenX 43% of the total. Of the other PFASs, PFPeA (4%) and PFHxA (4%) were detected in the highest concentrations. There is a large variety in the concentrations found. Highest concentration of PFOA (3020 pg/g ww) was found in a cucumber in a greenhouse (with high concentrations of PFOA in soil). GenX was found at a maximum of 5280 pg/g ww (endive, close to the FCPP). Maximum of PFPeA was 290 pg/g ww (tomato, close to the FCPP). For PFUnDA, the highest level was 160 pg/g ww in a tomato sample (upwind to the FCPP), it is unclear what caused this elevated PFUnDA concentration. For the other PFASs, the maximum (individual) concentrations were below 100 pg/g ww and of those, the highest PFOS concentration was 37.6 pg/g ww. The detection frequency (DF, i.e. values above LOQ) of the EFSA-4 compounds (PFOA, PFOS, PFNA and PFHxS) was 73%, 22%, 20% and 3% respectively. For GenX it was 38%. Other compounds with a higher DF were PFHpA, PFHxA and PFBS with a frequency of 34%, 41% and 31% respectively. The remaining compounds had a DF lower than 14%.

The predominance of PFOA and GenX in the samples was expected as these two compounds were emitted at the FCPP (Zeilmaker et al., 2016). In 2017, a selection of garden produce was analysed for PFOA and GenX (Mengelers et al., 2018). Of the 35 samples analysed, only 5 samples had concentrations above the LOQ. However, it must be remembered that in 2017, analytical LOQ's were much higher (1000 pg/g ww for GenX and PFOA) than they are now. The garden produce analysed in 2017 were mainly vegetables from gardens that were less than 1 km from the FCPP. The concentrations found in 2017 were in the same order of magnitude as in the current study (2017: 1000–3500 pg/g ww for GenX (n = 4) and 1700 pg/g ww PFOA (n = 1). 2021: 100–5600 pg/g ww GenX (n = 4), 800 pg/g ww PFOA (n = 1). In the current study, concentrations above 1000 pg/g ww were only measured at the locations closest to the FCPP (<1 km), while locations that were further away had lower concentrations. E.g. in the reference allotments average sum PFASs was 9 pg/g, which is comparable to the levels found in grocery stores when comparing to similar vegetables (Schepens et al., 2023).

3.2. PFASs levels relative to distance from the FCPP

The highest concentrations in produce are clearly observed at the locations closest to the FCPP (Fig. 2). Highest concentrations of GenX and PFOA were measured in the three individual gardens closest to the facility (≤1 km), and especially PFOA showed very high concentrations in a cucumber and tomato in a greenhouse, which was explained by high PFOA concentrations in the soil, specifically within this greenhouse. That such high PFOA concentrations are still found in the garden produce, even 9 years after PFOA was phased out (Zeilmaker et al., 2016), can be explained by the persistence of PFOA. In the gardens closest to the FCPP, the GenX concentrations are higher than PFOA concentrations. This is likely due to the fact that at the time of sampling, GenX was likely

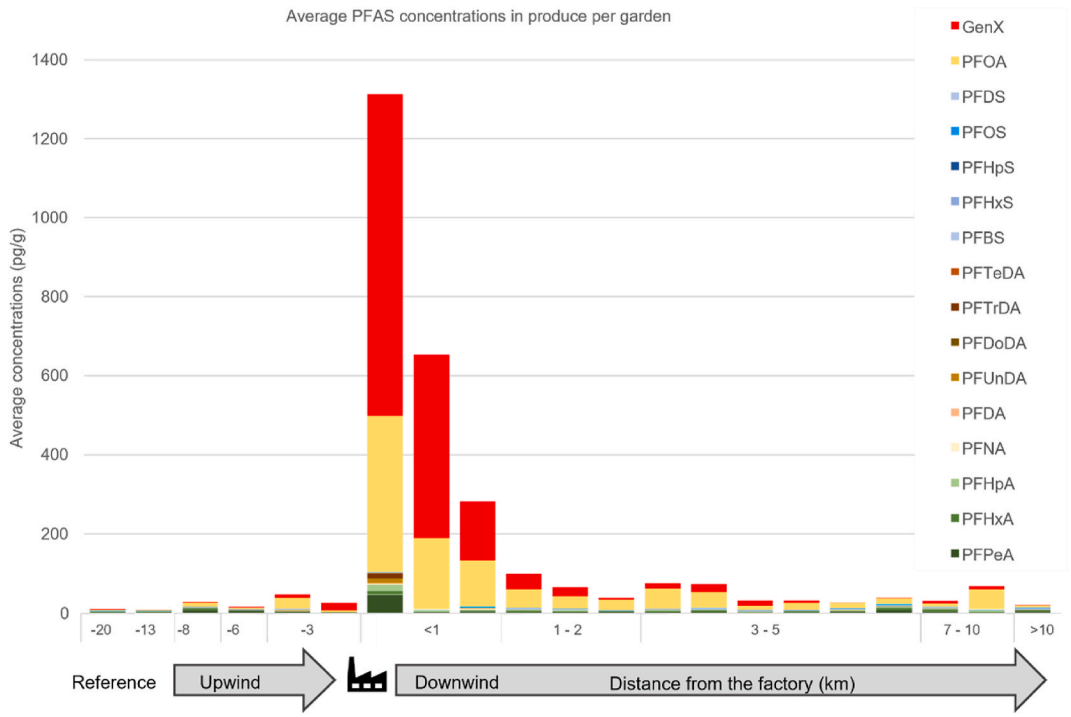


Fig. 2. Average PFASs concentrations in garden produce per allotment/garden. Distance from the FCPP is shown in km. PFASs concentrations below the limit of detection were set to 0 (lowerbound).

still emitted from the FCPP (the permit for GenX was set at 95 kg/year in 2020 and 4 kg/year in 2022 (DCMR, 2023)). The other PFASs detected (Fig. 2) were also found at highest concentrations closest to the FCPP, though at much lower levels than PFOA and GenX. Levels in the produce are also mirrored by concentrations in the soil and groundwater, where the highest levels in soil were also observed closest to the FCPP (supplemental information, Fig. S2) (van Bentum and Pancras, 2023).

Similar observations were made at other fluorochemical production factories around the world. In Huantai County, China, PFOA concentrations were found to be highest in soil and water samples close to and downwind from the fluoropolymer production facility (Liu et al., 2016). Liu et al. (2019) investigated uptake in crops close to a fluorochemical industrial park (FIP) in China (distance of 0.3 and 10 km from the FIP). The results reported were similar to these in Dordrecht in the sense that PFASs concentrations in surface waters were highest at sites closest to the FIP. They found ∑PFASs levels in edible parts of crops up to nearly 4000 ng/g dw at 0.3 km from the FIP, whereas the levels at 10 km were much lower (up to 30 ng/g dw). It should be noted that the levels reported in their study (and some other studies cited below) were expressed on ng/g dw, contrary to our study (pg/g ww), meaning that their levels were orders of magnitude higher than in the current study. In later research, Liu et al. (2022) analysed PFASs in crops, soil and surface water samples taken within a 20 km radius around a FIP in Changshu, China. The FIP consists of several fluorochemical facilities either producing or applying perfluoroalkyl acids, and in the close vicinity there are also textile and apparel factories, metal processing plants and electronics manufacturers. However, the spatial distribution of PFASs in soil did not show a significant decrease with increasing distance from the FCPP, which could be ascribed to the presence of other PFASs sources. Sum PFASs concentrations in rice grains ranged from 6.19 to 143.13 ng/g dw. Liu et al. found that legacy PFOA and PFOS were still present (levels ranging from 0.12 to 1.91 ng/g dw for PFOS and from not detected–0.84 ng/g dw for PFOA), but shorter-chained PFASs (like PFBA) and novel/replacement PFASs like (GenX and 6:2 FTS) were found in higher concentrations at some locations (Liu et al., 2016). Nearby these fluorochemical producers, high concentrations of PFBA, PFPeA, PFHxA and GenX were detected in soil. Galloway et al. (2020), found PFOA and GenX in soils and water up to 28 km downwind a fluoropolymer production facility in Parkersburg, USA. The presence of PFASs in crops was not evaluated in the study.

3.3. PFASs concentrations by type of garden produce

Fig. 3 shows the average concentrations by produce category from the studied allotments (Fig. 3 top) and from the reference allotments (Fig. 3, bottom). The three individual gardens directly downwind of the FCPP (<1 km) were excluded from Fig. 3 (top) because the number and type of fruits and vegetables samples were limited whereas the levels were much higher than in the other allotments. This would have hampered a good comparison among garden produce samples. Data from these gardens is given in SI – Fig. S2. The results (Fig. 3) show that the highest average PFOA concentrations were found in root vegetables and the lowest concentrations in fruit and bulb vegetables (onion). Average concentrations of GenX were almost similar for 5 of the 9 categories (10 pg/g ww) and much lower in the other 4 categories (1–2 pg/g ww). Overall, highest PFASs concentrations were detected in brassicas, leafy vegetables and root vegetables, as will be discussed below. Minimum, maximum and average concentrations for PFOA and GenX per type of fruit and vegetable are given in SI – Tables S4 and S5. Data from other PFAS can be found in Boon and te Biesebeek (2022). The high concentrations of PFOA and GenX resulted in an advice to omit consumption of the fruit and vegetables of the gardens closest (≤1 km) to the FCPP and to limit consumption from the gardens 1–10 km lateral and downwind the FCPP (Boon and te Biesebeek, 2022). Although different vegetables show different uptake of PFASs, these differences did not lead to advice on the consumption of specific vegetables (Boon and te Biesebeek, 2022).

As regards the reference allotments (Fig. 3, bottom), the average summed PFASs concentrations across all fruits and vegetables was 9 pg/g ww. In another study in the Netherlands on the dietary exposure of the general Dutch population (i.e. not related to a point source), fruits and vegetables from supermarkets, week markets and grocery stores were

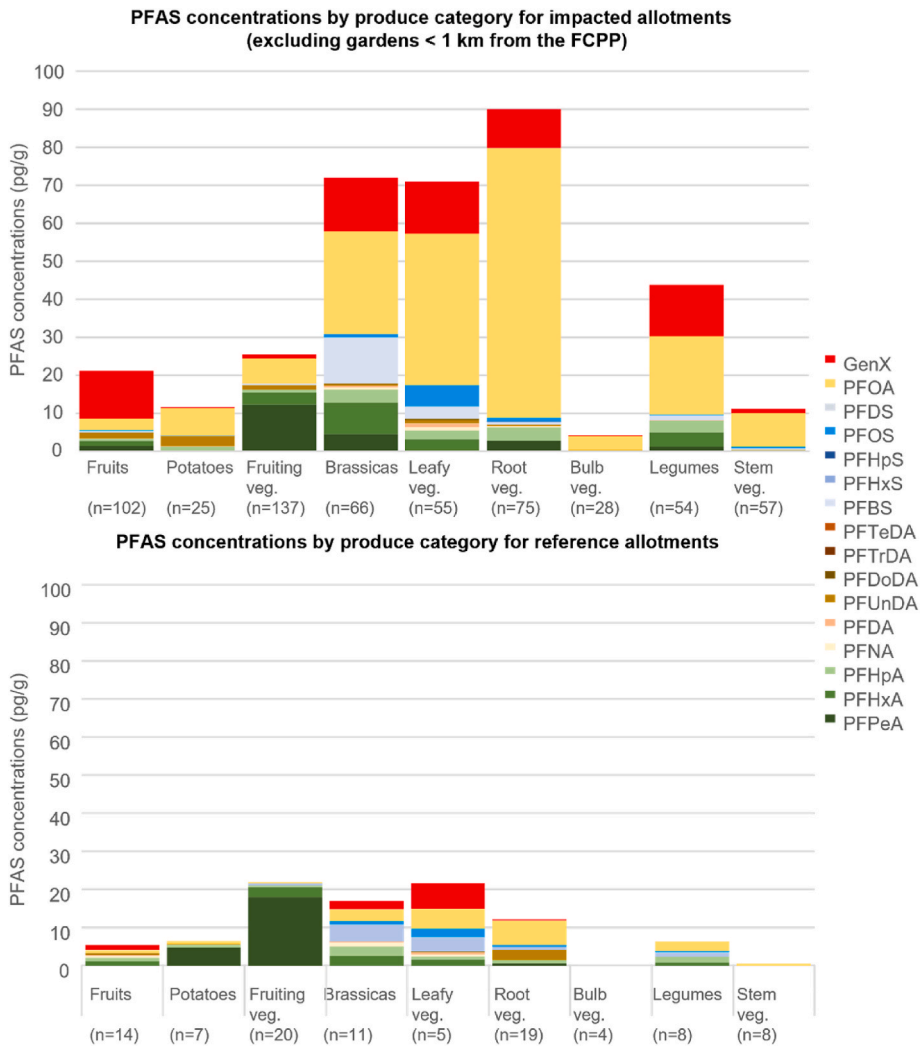


Fig. 3. Average PFAS concentrations found in different types of garden produce. Averages include all samples within the respective category. Concentrations below the LOD were set to 0. Top: all allotments (except for gardens <1 km from the FCPP and the reference allotments). Bottom: Reference allotments. Veg. = vegetables.

analysed with the same analytical method. The levels found in the current study are comparable to the average summed PFASs concentrations of 7 pg/g ww, when comparing similar vegetables (Schepens et al., 2023).

Looking closer to individual garden produce (instead of categories as above), some observations can be made. Specifically, for PFOA and GenX in SI-tables S4 and S5, data is presented per fruit and vegetable type. Looking at the majority of the impacted allotments (1–14 km) for PFOA, root vegetables (beetroot, carrot) showed higher average levels (103 and 55 pg/g ww resp.). The same holds for the average of leafy vegetables (chard (1 observation), spinach, endive and lettuce with 335, 55, 37 and 30 pg/g ww, resp.) as well as the average of 1 type of the brassicas (kale, 46 pg/g ww). Several papers (Blaine et al., 2014; Costello and Lee, 2020; Lesmeister et al., 2021; Mei et al., 2021) critically discussed or reviewed the uptake and translocation of PFASs in crops. They mentioned several factors impacting the uptake and translocation. A stronger developed Casparian strip is a first barrier for longer chain compounds to enter the xylem, meaning that these long chain PFASs will stay in the roots rather than the above-ground parts. A lesser developed Casparian strip allows translocation with xylem to the shoots, which may explain the higher levels observed in leafy vegetables. Once PFAS are translocated to the leaves, they may be translocated further to other parts (e.g. fruits) of the plant through the phloem, although this may require overcoming another barrier (Lesmeister et al., 2021). This may partially explain why little PFOA or GenX was transported to the fruits which are dependent on transport via the phloem (Blaine et al., 2014). Other factors may also play a role such as higher plant transpiration rates, resulting in larger translocation from roots to shoots, passive and active transport (e.g. through protein-carriers). This results in a complex interplay of these factors that determine the uptake in plants.

The variation of GenX levels (SI-table S5) for various fruits and vegetables was less pronounced, compared to PFOA. There is limited data available on GenX bioaccumulation by plants (Costello and Lee, 2020; Lesmeister et al., 2021). GenX is considered to be a mobile compound in the environment (Yan et al., 2020) and possibly the high water solubility of this compound facilitates the uptake of GenX in some of the plants with higher transpiration rates (e.g. lettuce, endive).

Next to direct uptake from soil, garden produce may be contaminated from atmospheric deposition from the FCPP. As GenX was still emitted from the FCPP in the year of sampling, this may have played a role for the contamination of e.g. leafy vegetables. But not for PFOA as emissions of PFOA were stopped 9 years before this study started. That rules out PFOA leaf uptake directly from emissions, although leaf uptake from deposition of PFAS-contaminated dust or soil particles cannot be ruled out. Leafy vegetables with large leaf surface area and extended growing time (e.g. kale) may be more susceptible to aerial deposition, which may explain the elevated average level.

Also for other PFASs (other than PFOA and GenX, data not shown)

some observations were made. Tomatoes showed a higher levels of PFHxA (average 16 pg/g ww (n = 29), range 2–77 pg/g ww, detection frequency (DF) = 81%), PFHpA (average 5.3 pg/g ww (n = 34), range 0.8–69 pg/g ww, DF = 94%) compared to other compounds. Concentrations in Brussels sprouts were the lowest of all produce and the PFASs profile was almost exclusively PFOA, suggesting that other PFASs are hardly taken up from the soil by these plants. There were no clear differences for leafy vegetables (endive, spinach, lettuce and chard) where PFBS (average 13 pg/g ww (n = 82), range 0.5–74 pg/g ww, DF = 74%), PFHxA (average 7.9 pg/g ww (n = 101), range 0.7–81 pg/g ww, DF = 91%) and PFHpA (average 5.2 pg/g ww, range 0.6–25 pg/g ww, DF = 80%) were frequently detected (though at lower levels than PFOA and GenX). In legumes, PFHxA (average 17 pg/g ww (n = 14), range 2.5–21 pg/g ww, DF = 38%) and PFHpA (average 2.6 pg/g ww (n = 37), range 0.6–7 pg/g ww, DF = 100%) were frequently detected (next to PFOA and GenX). The contamination profile for strawberries, apples and pears did not show dominance of any particular PFASs. As the FCPP predominantly emitted PFOA and GenX, the source of the other PFASs found in these fruits and vegetables is less clear. Some of the PFAS (e.g. PFHpA, PFHxA) might originate from impurities in the PFOA used, other possible sources may be other nearby industrial activities (e.g. waste incinerator), not yet characterised emissions from the FCPP and disperse pollution sources (e.g. wet and dry deposition) but this requires further investigations. The data of the garden produce of the reference gardens also shows that there is a certain ambient background level of PFASs, likely coming from disperse (not characterised) sources. Lasters et al. (2024) recently published a study on PFASs in fruits and vegetables close to a Belgian point source (Antwerp region). While soil in that area is known to be contaminated with per- and polyfluoralkyl sulfonic acids (PFSAs), they hardly detected PFOS and PFBS in the produce, which they related to the stronger sorption of these PFSAs to soil and consequently a lower bioavailability for uptake. Next to that, their analytical method had higher LOQs compared to the current study, which may have resulted in larger number of non-detects. They also analysed a broad suite of perfluorinated carboxylic acids (PFCAs) and found for the PFCAs and 6:2 fluorotelomer sulfonate highest levels in legumes > leafy vegetables ≈ fruiting vegetables > herbs > walnut > root vegetables ≈ shoot vegetables ≈ large/small fruit vegetables. While in our study root vegetables, leafy vegetables and brassicas showed the highest values, this may be due to a different suite of PFASs included in both studies as well as different selection of fruits and vegetables. Their ∑PFAS-levels were in the 1.5–5 ng/g ww range, being 2-3 orders of magnitude higher than in the current study.

3.4. Relationship between PFASs concentrations in garden produce and soil

The highest PFASs levels in soil were detected in the private gardens directly downwind of the FCPP, especially in a greenhouse in one of the gardens. Lower levels of PFASs were found in soils upwind (the reference locations) and further downwind from the FCPP. Levels are shown in Fig. S3. The concentrations of PFOA were mostly lower in the topsoil than in the deeper soil layer (van Bentum and Pancras, 2023) indicating downward transport of historically deposited PFOA. The transport to deeper layers has also been shown in this area by Gerardu et al. (2023).

Linear regression was used to screen the most relevant environmental concentrations and conditions to further evaluate the relationships with concentrations in produce. A substantial linear regression with PFOA concentrations in the topsoil (0–20 cm) was specifically found for all legumes, rooting vegetables and bulb vegetables (Table S6). The concentration of PFOA in other garden produce types could only be poorly explained by linear regression with the PFOA concentration in the topsoil, which can be related to heterogeneous field circumstances. The concentrations of PFOA in irrigation water, the concentrations of PFOA in the deeper soil layer or the soil properties could not explain the observed concentration of PFOA in produce (van Bentum and Pancras, 2023). Environmental concentrations of the other PFASs were mostly below the LOD and relationships could not be assessed. Hence, only the concentrations of PFOA and GenX in garden produce in relation to concentrations in the topsoil are further discussed here.

PFASs bioaccumulation factors (BAFs) were calculated for all samples as PFASs concentration in garden produce (pg/g ww) divided by the PFASs concentration in topsoil (ng/kg dw). Since all samples were analysed based on fresh weight for the risk assessment, BAFs were calculated based on fresh weight and then converted to BAFs based on dry weight (dw) using average produce dry weight data obtained from the Dutch food table (RIVM, 2023). The PFOA and GenX BAFs per produce type are given in Fig. 4 based on fresh weight (left) and dry weight (right). The bioaccumulation factors based on dry weight are generally 10-fold higher than those for wet weight. The difference is larger for leafy vegetables, since these vegetables have a relatively high moisture content.

In this study, the BAFs (dw basis) for PFOA were the highest for leafy vegetables and rooting vegetables and lowest for the fruits. BAFs for GenX (Table S7) could only be derived for 67 produce samples as other GenX concentrations were below the LOD in the topsoil (<0.1 μg/kg dw). BAFs for GenX (dw basis) were highest in leafy vegetables, followed by legumes and rooting vegetables, illustrating that the same types of vegetables had the highest BAFs as for PFOA, although in a different order. For GenX, the leafy vegetables clearly show the highest BAFs. This might be due to ongoing deposition during the growing season of 2021. Factors explaining higher uptake and translocation, and therefore higher levels in edible parts have been discussed above. These generally lead to higher BAF values for these fruits and vegetables. As the atmospheric emission of PFOA was halted at the time of sampling the direct uptake by leaves from emissions can no longer explain different produce concentrations. However, the higher BAFs of leafy vegetables and brassicas may still be related to the deposition of contaminated soil and dust on the leaves as previously shown for other contaminants (Engel-Di Mauro, 2021).

The BAFs for PFOA (dw basis) found in this study were generally 1 order of magnitude lower than those found in review papers of crop PFASs accumulation Ghisi et al. (2019); Costello and Lee (2020); Lesmeister et al. (2021); Xu et al. (2022); Lesmeister et al. (2021) reported a median dw PFOA BAF of approximately 0.3, while the median BAF for PFOA in this study was 0.03. The conversion from fresh to dry weight may have caused some uncertainty, but is unlikely to account for this difference. There may be several explanations for differences in our study compared to other studies. The time period since PFOA emissions ceased from the FCPP may play an important role in the sequestration of PFOA and thus make it less bioavailable for uptake by plants. Other organic pollutants have been shown to become less bioavailable as they age (Alexander, 2000). Cultivation and leaching via rainfall (Gerardu et al., 2023) may have washed PFOA out from the top layer to deeper layers, resulting in less bioavailable PFOA remaining. Knight et al. (2021) found little to no aging effect for PFOA over a six-month period under controlled conditions. However, six months is very short in comparison to the time scale here for which the PFOA has been aged and experimental data investigating the long-term aging PFASs in the field is a clear current knowledge gap (Naidu et al., 2020; Sima and Jaffé, 2021). In this study, growing conditions were not controlled (nor were they the same for each allotment or garden) as produce was grown by citizens and this may have additionally influenced the BAFs. Nevertheless, field circumstances have a better resemblance to actual exposure to humans than lab circumstances. Evidence of similar effects was reported previously as the BAF of PFOA in lettuce was reported to be lower in a field experiment than in a controlled greenhouse experiment (Blaine et al., 2013). BAFs have also been reported to be higher when PFASs were present in irrigation water rather than naturally present in soil (Blaine et al., 2013; Bao et al., 2020; Gredelj et al., 2020). At the investigated allotments in this study, PFASs concentrations in irrigation water (mainly rain water) were in the low ng/l level, and therefore not

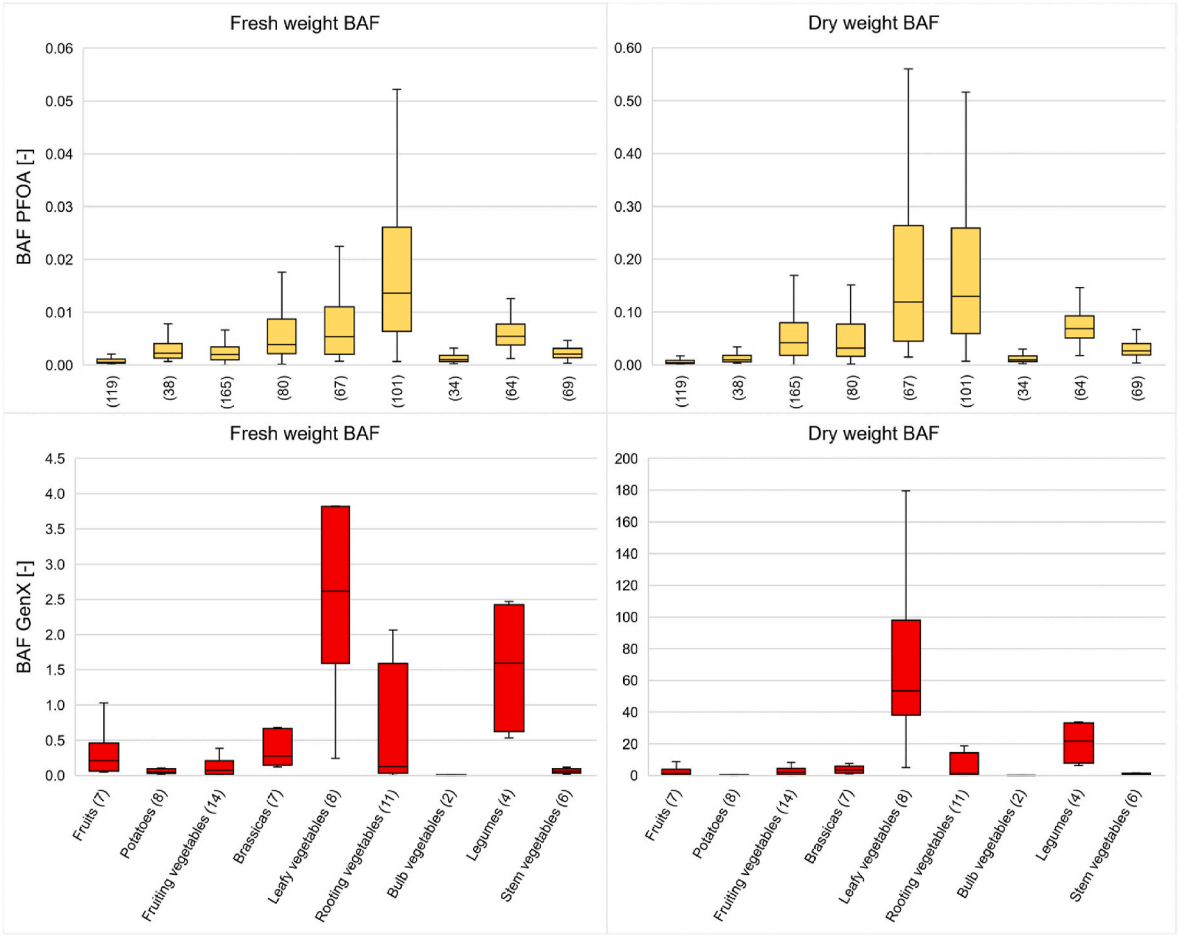


Fig. 4. Bioaccumulation factors (BAF) for different types of produce for PFOA (top) and GenX (bottom). BAFs were derived based on fresh weight (left) and converted to dry weight (right). Number of observations are given between brackets. Outliers are not shown.

reported here (van Bentum and Pancras, 2023). A combination of these aforementioned processes and aging phenomena may explain the relatively low field derived BAFs for PFOA.

4. Summary and conclusion

This research investigated PFASs concentrations in 28 types of garden produce from 21 allotments and gardens that were located in a large area around a FCPP in the Netherlands. The concentrations of PFASs in the garden produce were highest directly downwind of the FCPP and decreased with increasing distance. Although the emission of PFOA ceased in 2012, elevated levels of this PFASs were still detected 9 years later in the garden produce of the individual gardens and allotments up to 10 km downwind from the FCPP, compared to the reference allotments. In the private gardens directly downwind the FCPP, GenX was measured at higher levels than PFOA, which could be attributed to the more recent (and ongoing) emissions.

The levels of PFASs (mainly PFOA) were highest in brassicas, leafy vegetables and root vegetables. The PFASs levels in garden produce and the PFASs levels in corresponding soils were used to calculate BAFs. The values were approximately 1 order of magnitude lower than the bioconcentration factors previously reported from other locations. This may be due to aging of the contamination and the intense cultivation of the garden plots which reduced the bioavailability of the PFASs contamination.

More study is required on BAFs under field circumstances, as BAFs are one of the most important factors in the derivation of soil risk levels. A significant overestimation of BAFs will lead to risk levels that are stricter than necessary, which may lead to more expensive mitigating measures than necessary.

The elevated PFASs concentrations in the soil and garden produce from the area surrounding the FCPP highlight the need for further site investigations and risk assessments when a new allotment garden is started.

CRediT authorship contribution statement

Tessa Pancras: Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Elisabeth van Bentum:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Leontien de Pagter:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Investigation, Funding acquisition. **Maarten van Hoef:** Writing – review & editing, Writing – original draft, Investigation. **Ron Hoogenboom:** Writing – review & editing, Writing – original draft, Investigation. **Bjorn Berendsen:** Writing – review & editing, Writing – original draft, Validation, Supervision, Investigation, Data curation. **Stefan P.J. van Leeuwen:** Writing – review & editing, Writing – original draft, Supervision, Resources, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.chemosphere.2024.143651.

Data availability

Data will be made available on request.

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